Oxygenation

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Outlines

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- Process of oxygenation
- Regulation of respiration
- Factors Affecting Oxygenation
- Alterations in Respiratory Functioning
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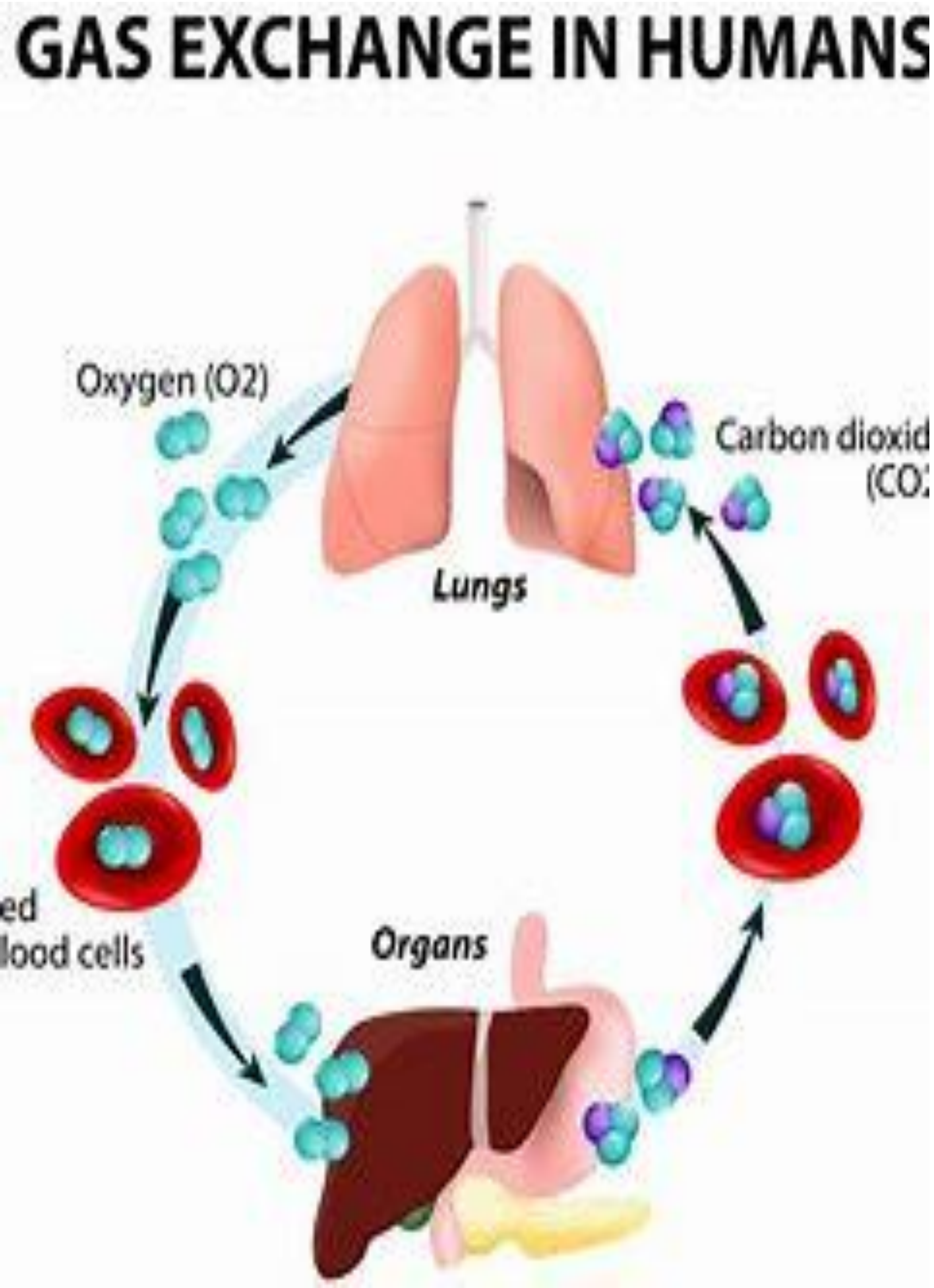
Introduction

- Oxygen is necessary to sustain life. The cardiac and respiratory systems supply the body with oxygen. Blood is oxygenated through the mechanisms of ventilation, perfusion, and transport of respiratory gases. Neural and chemical regulators control the rate and depth of respiration in response to changing tissue oxygen demands.

Cont, Introduction

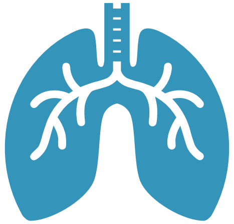
- Oxygen, a clear, odorless gas that constitutes approximately 21% of the air we breathe, is necessary for proper functioning of all living cells.
- The absence of oxygen can lead to cellular, tissue, and organ death. Cellular metabolism produces carbon dioxide, which must be eliminated from the body to maintain normal acid–base balance.



Physiology of Respiratory System

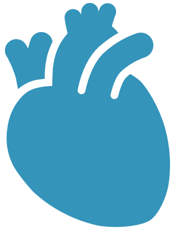
The exchange of respiratory gases occurs between the environment and the blood. Respiration is the exchange of oxygen and carbon dioxide during cellular metabolism. The airways of the lung transfer oxygen from the atmosphere to the alveoli, where the oxygen is exchanged for carbon dioxide. Through the alveolar capillary membrane, oxygen transfers to the blood, and carbon dioxide transfers from the blood to the alveoli.

There are three steps in the process of oxygenation: ventilation, perfusion, and diffusion.



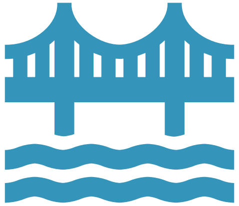
Ventilation

is the process of moving gases into and out of the lungs. It requires coordination of the muscular and elastic properties of the lung and thorax.



Perfusion

Is the ability of the cardiovascular system to pump oxygenated blood to the tissues and return deoxygenated blood to the lungs.



Diffusion

The process of moving the respiratory gases from one area to another by concentration gradients..

Work of Breathing: (WOB) is the effort required to expand and contract the lungs. In the healthy individual breathing is quiet and accomplished with minimal effort. The amount of energy expended on breathing depends on the rate and depth of breathing, the ease in which the lungs can be expanded (compliance), and airway resistance.

Inspiration is an active process, stimulated by chemical receptors in the aorta.

Expiration is a passive process that depends on the elastic recoil properties of the lungs, requiring little or no muscle work.

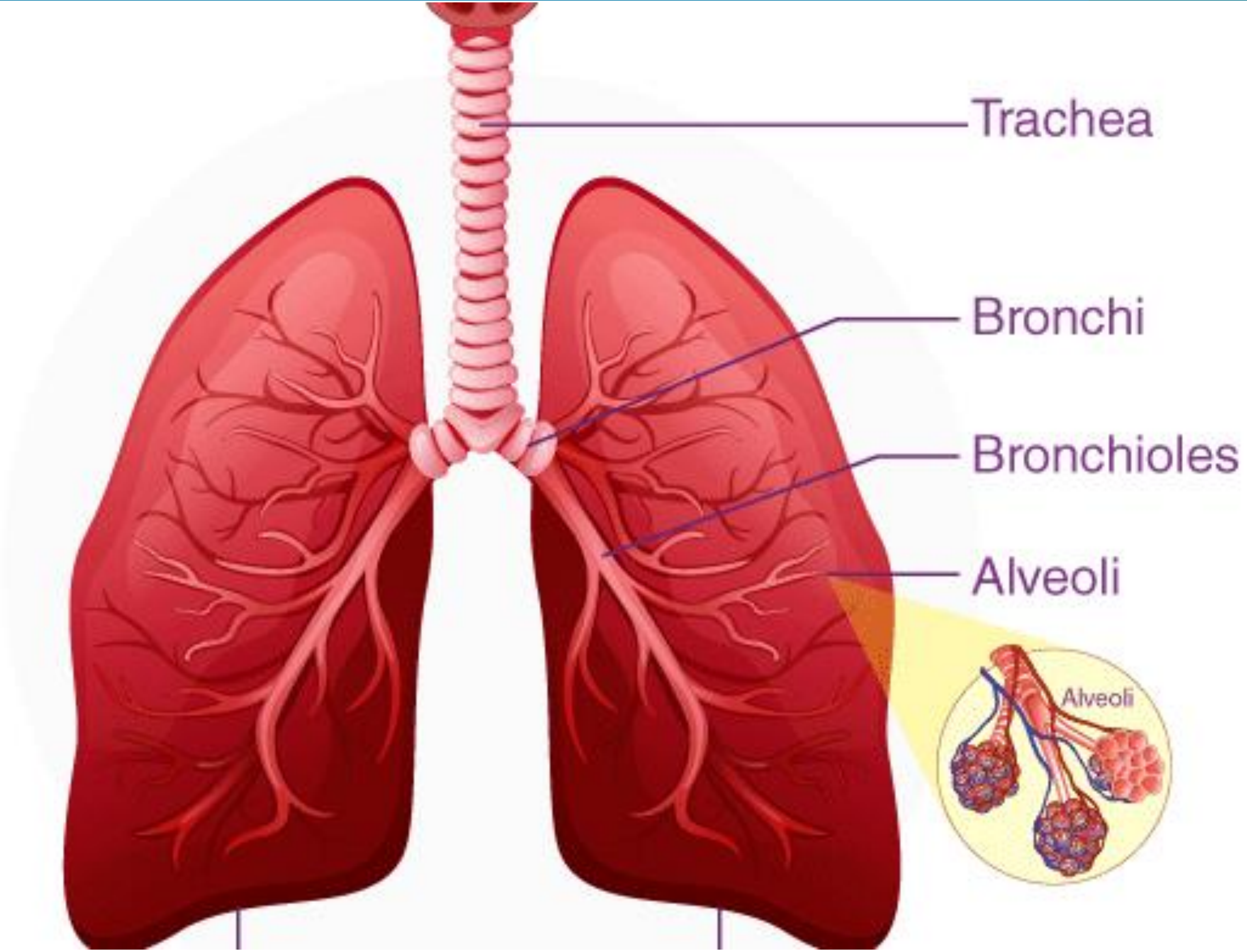
Surfactant is a chemical produced in the lungs to maintain the surface tension of the alveoli and keep them from collapsing. Patients with advanced chronic obstructive pulmonary disease (COPD) lose the elastic recoil of the lungs and thorax. As a result, the patient's work of breathing increases. In addition, patients with certain pulmonary diseases have decreased surfactant production and sometimes develop atelectasis. (***Atelectasis*** *is a collapse of the alveoli that prevents normal exchange of oxygen and carbon dioxide.*)

Compliance is the ability of the lungs to distend or expand in response to increased intra-alveolar pressure.

- Compliance decreases in diseases such as pulmonary edema, interstitial and pleural fibrosis, and congenital or traumatic structural abnormalities such as kyphosis or fractured ribs.

- Decreased lung compliance, increased airway resistance, and the increased use of accessory muscles increase the WOB, resulting in increased energy expenditure. Therefore the body increases its metabolic rate and the need for more oxygen. The need for elimination of carbon dioxide also increases causing further deterioration of respiratory status.

- ***Tidal volume*** is the amount of air exhaled after normal inspiration.
- ***Residual volume*** is the amount of air left in the alveoli after a full expiration.
- ***Forced vital capacity*** is the maximum amount of air that can be removed from the lungs during forced expiration.


Trachea
Bronchi
Bronchioles
Alveoli
Alveoli

Regulation of Respiration

- Regulation of respiration is necessary to ensure sufficient oxygen intake and carbon dioxide elimination to meet the demands of the body (e.g., during exercise, infection, or pregnancy).

- Neural and chemical regulators control the process of respiration.

  - Neural regulation includes the central nervous system control of respiratory rate, depth, and rhythm. The cerebral cortex regulates the voluntary control of respiration by delivering impulses to the respiratory motor neurons by way of the spinal cord.

- Chemical regulation maintains the appropriate rate and depth of respirations based on changes in the carbon dioxide (CO_2), oxygen (O_2), and hydrogen ion (H^+) concentration (pH) in the blood. Changes in chemical content of O_2, CO_2, and H (pH) stimulate the chemoreceptors located in the medulla, aortic body, and carotid body, which in turn stimulate neural regulators to adjust the rate and depth of ventilation to maintain normal arterial blood gas levels.

Factors Affecting Oxygenation

- **Five factors** influence adequacy of circulation, ventilation, perfusion, and transport of respiratory gases to the tissues:

(1) **Physiological**

(2) **Conditions Affecting Chest Wall Movement**

(3) **Developmental**

(4) **Lifestyle**

(5) **Environmental.**

I Physiological Factors

1. **Any condition affecting cardiopulmonary functioning** directly affects the ability of the body to meet oxygen demands. Respiratory disorders include hyperventilation, hypoventilation, and hypoxia. Cardiac disorders as impaired valvular function, myocardial hypoxia,and peripheral tissue hypoxia.

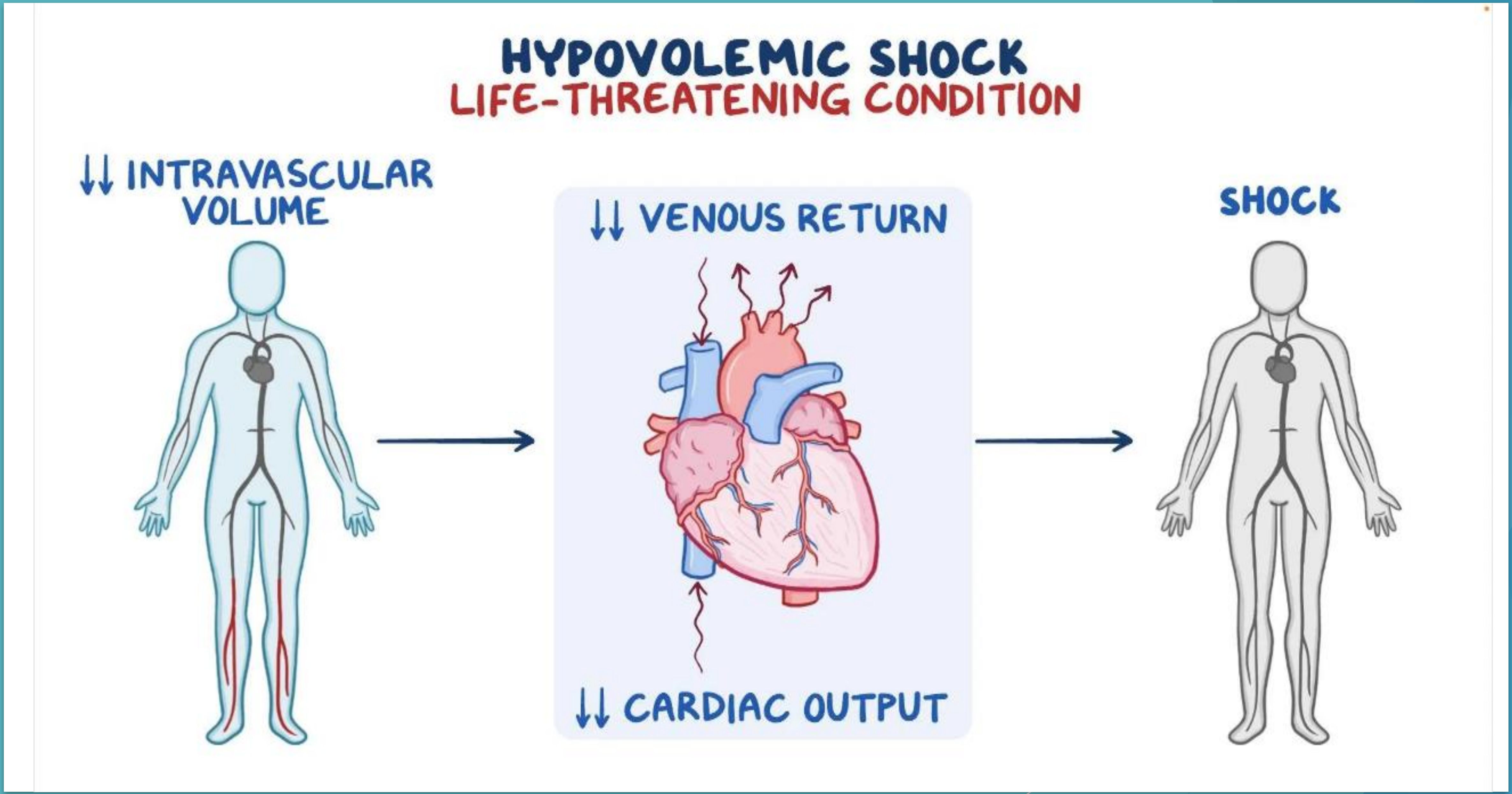
2. **Decreased Oxygen-Carrying Capacity**

Hemoglobin carries the majority of oxygen to tissues. Anemia and inhalation of toxic substances decrease the oxygen-carrying capacity of blood by reducing the amount of available hemoglobin to transport oxygen. Anemia (i.e., a lower-than-normal hemoglobin level) is a result of decreased hemoglobin production, increased red blood cell destruction, and/or blood loss. Oxygenation decreases as a secondary effect with anemia. The physiological response to chronic hypoxemia is the development of increased red blood cells (polycythemia).

٣.Hypovolemia

- ✓ Conditions such as shock and severe dehydration cause extracellular fluid loss and reduced circulating blood volume, or **hypovolemia.**

Decreased circulating blood volume results in hypoxia to body tissues. With significant fluid loss, the body tries to adapt by peripheral vasoconstriction and increasing the heart rate to increase the volume of blood returned to the heart, thus increasing the cardiac output.



HYPOVOLEMIC SHOCK
LIFE-THREATENING CONDITION
↓↓ INTRAVASCULAR VOLUME
↓↓ VENOUS RETURN
↓↓ CARDIAC OUTPUT
SHOCK

٤ . Decreased Inspired Oxygen Concentration

- Decreases in the fraction of inspired oxygen concentration (FiO2) are caused by upper or lower airway obstruction, which limits delivery of inspired oxygen to alveoli; decreased environmental oxygen (at high altitudes); or hypoventilation (occurs in drug overdoses).

٥. Increased Metabolic Rate.

Increased metabolic activity increases oxygen demand. The level of oxygenation declines when body systems are unable to meet this demand. An increased metabolic rate is normal in pregnancy, wound healing, and exercise because the body is using energy or building tissue.

II. Conditions Affecting Chest Wall Movement

- Any condition reducing chest wall movement results in decreased ventilation. If the diaphragm does not fully descend with breathing, the volume of inspired air decreases, delivering less oxygen to the alveoli and tissues.

1. Pregnancy.

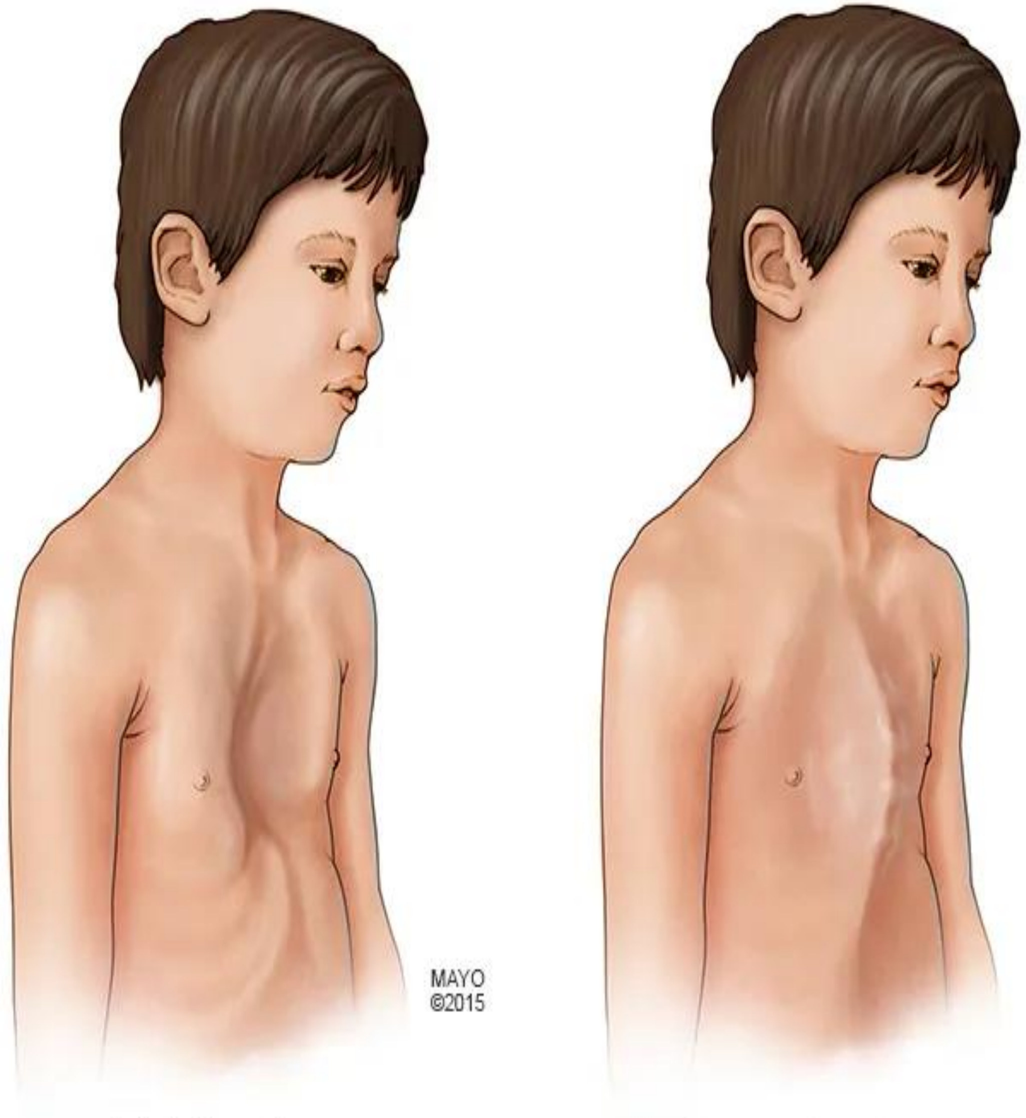
- As the fetus grows during pregnancy, the enlarging uterus pushes abdominal contents upward against the diaphragm. In the last trimester of pregnancy, the inspiratory capacity declines, resulting in dyspnea on exertion and increased fatigue.

2. Obesity

- Patients who are morbidly obese have reduced lung volumes from the heavy lower thorax and abdomen, particularly when in the recumbent and supine positions.
- Morbidly obese patients have a reduction in lung and chest wall compliance as a result of entrance of the abdomen into the chest, increased WOB, and decreased lung volumes.
- The obese patient is also susceptible to atelectasis or pneumonia after surgery because the lungs do not expand fully and the lower lobes retain pulmonary secretions.

٣. Musculoskeletal Abnormalities.

- Musculoskeletal impairments in the thoracic region reduce oxygenation. Such impairments result from abnormal structural configurations, trauma, muscular diseases, and diseases of the central nervous system. Abnormal structural configurations impairing oxygenation include those affecting the rib cage such as pectus excavatum and the vertebral column such as kyphosis, lordosis, or scoliosis.



Indented breastbone (pectus excavatum)

Protruding breastbone (pectus carinatum)

4. Trauma.

- Flail chest is a condition in which multiple rib fractures cause instability in part of the chest wall. The unstable chest wall allows the lung underlying the injured area to contract on inspiration and bulge on expiration, resulting in hypoxia.
- Patients with thoracic or upper abdominal surgical incisions use shallow respirations to avoid pain, which also decreases chest wall movement. Opioids used to treat pain depress the respiratory center, further decreasing respiratory rate and chest wall expansion.

5. Neuromuscular Diseases.

- Neuromuscular diseases affect tissue oxygenation by decreasing the patient's ability to expand and contract the chest wall. Ventilation is impaired, resulting in atelectasis, hypercapnia, and hypoxemia. Examples of conditions causing hypoventilation include poliomyelitis.

6. Central Nervous System Alterations.

- Diseases or trauma of the medulla oblongata and/or spinal cord result in impaired respiration. When the medulla oblongata is affected, neural regulation of respiration is impaired, and abnormal breathing patterns develop.

7. Influences of Chronic Disease.

- Oxygenation decreases as a direct consequence of chronic lung disease. Changes in the anteroposterior diameter of the chest wall (barrel chest) occur because of overuse of accessory muscles and air trapping in emphysema.

III-Developmental Factors:

- The developmental stage of a patient and the normal aging process affect tissue oxygenation.

1. Infants and Toddlers:

- Infants and toddlers are at risk for upper respiratory tract infections as a result of frequent exposure to other children, an immature immune system, and exposure to secondhand smoke. In addition, during the teething process some infants develop nasal congestion, which encourages bacterial growth and increases the potential for respiratory tract infection.

2. School-Age Children and Adolescents:

- School-age children and adolescents are exposed to respiratory infections and respiratory risk factors such as cigarette smoking or secondhand smoke. A healthy child usually does not have adverse pulmonary effects from respiratory infections.
- A person who starts smoking in adolescence and continues to smoke into middle age has an increased risk for cardiopulmonary disease and lung cancer.

3. Young and Middle-Aged Adults:

- Young and middle-aged adults are exposed to multiple cardiopulmonary risk factors: an unhealthy diet, lack of exercise, stress, over-the-counter and prescription drugs not used as intended, illegal substances, and smoking.

4. Older Adults:

- The cardiac and respiratory systems undergo changes throughout the aging process. The changes are associated with calcification of the heart valves, and costal cartilages. The arterial system develops atherosclerotic plaques. Osteoporosis leads to changes in the size and shape of the thorax. The trachea and large bronchi become enlarged from calcification of the airways. The alveoli enlarge, decreasing the surface area available for gas exchange. The number of functional cilia is reduced, causing a decrease in the effectiveness of the cough mechanism, putting the older adult at increased risk for respiratory infections.

IV-Lifestyle Factors

- Lifestyle modifications are difficult for patients because they often have to change an enjoyable habit such as cigarette smoking or eating certain foods.
- Risk-factor modification is important and includes smoking cessation, weight reduction, a low-cholesterol and low-sodium diet, management of hypertension, and moderate exercise. Although it is difficult to change long-term behavior, helping patients acquire healthy behaviors reduces the risk for or slows or halts the progression of cardiopulmonary diseases.

1. Nutrition:

- Nutrition affects cardiopulmonary function in several ways. Severe obesity decreases lung expansion, and increased body weight increases tissue oxygen demands.
- The malnourished patient experiences respiratory muscle wasting, resulting in decreased muscle strength. Cough efficiency is reduced secondary to respiratory muscle weakness, putting the patient at risk for retention of pulmonary secretions.
- Patients who are morbidly obese and/or malnourished are at risk for anemia.
- Dietary practices also influence the prevalence of cardiovascular diseases. Cardio protective nutrition includes diets rich in fiber; whole grains; fresh fruits and vegetables; nuts; antioxidants; lean meats, fish, and chicken; and omega-3 fatty acids.

2. Exercise:

- Exercise increases the metabolic activity and oxygen demand of the body. The rate and depth of respiration increase, enabling the person to inhale more oxygen and exhale excess carbon dioxide.
- People who exercise for 30 to 60 minutes daily have a lower pulse rate and blood pressure, decreased cholesterol level, increased blood flow, and greater oxygen extraction by working muscles.

3. Smoking:

- Cigarette smoking and secondhand smoke are associated with a number of diseases, including heart disease, COPD, and lung cancer.
- Cigarette smoking worsens peripheral vascular and coronary artery diseases. Inhaled nicotine causes vasoconstriction of peripheral and coronary blood vessels, increasing blood pressure and decreasing blood flow to peripheral vessels.
- The risk of lung cancer is 10 times greater for a person who smokes than for a nonsmoker.
- Exposure to environmental tobacco smoke (secondhand smoke) increases the risk of lung cancer and cardiovascular disease in the nonsmoker.

4. Substance Abuse:

- Excessive use of alcohol and other drugs impairs tissue oxygenation in two ways;
  - **First,** the person who chronically abuses substances often has a poor nutritional intake. With the resultant decrease in intake of iron-rich foods, hemoglobin production declines.
  - **Second**, excessive use of alcohol and certain other drugs depresses the respiratory center, reducing the rate and depth of respiration and the amount of inhaled oxygen.

5. Stress:

- A continuous state of stress or severe anxiety increases the metabolic rate and oxygen demand of the body. The body responds to anxiety and other stresses with an increased rate and depth of respiration.
- Most people adapt; but some, particularly those with chronic illnesses or acute life-threatening illnesses such as an MI, cannot tolerate the oxygen demands associated with anxiety.

IV-Environmental Factors

- The environment also influences oxygenation.
- The incidence of pulmonary disease is higher in urban areas than in rural areas.
- A patient's workplace sometimes increases the risk for pulmonary disease. Occupational pollutants include asbestos, dust, and airborne fibers.
- Asbestosis is an occupational lung disease that develops after exposure to asbestos. The lung with asbestosis often has diffuse interstitial fibrosis, creating a restrictive lung disease. Patients exposed to asbestos are at risk for developing lung cancer, and this risk increases with exposure to tobacco smoke.

Alterations in Respiratory Functioning

- Illnesses and conditions affecting ventilation or oxygen transport cause alterations in respiratory functioning. ***The three primary alterations are <u>hypoventilation, hyperventilation</u>, and <u>hypoxia</u>.***
- The goal of ventilation is to produce a normal arterial carbon dioxide tension ($PaCO_2$) between 35 and 45 mm Hg and a normal arterial oxygen tension (PaO_2) between 80 and 100 mm Hg.
- Hypoventilation and hyperventilation are often determined by arterial blood gas analysis.
- Hypoxemia refers to a decrease in the amount of arterial oxygen.
- Nurses monitor arterial oxygen saturation (SpO_2) using a noninvasive oxygen saturation monitor pulse oximeter. Normally SpO_2 is greater than or equal to 95%.

Alterations in Respiratory Functioning

The three primary alterations are

- ***Hypoventilation***
- ***Hyperventilation***
- ***Hypoxia.***
- **Cyanosis**

I- Hypoventilation:

- Hypoventilation occurs when alveolar ventilation is inadequate to meet the oxygen demand of the body or eliminate sufficient carbon dioxide.
- As alveolar ventilation decreases, the body retains carbon dioxide. For example, atelectasis, a collapse of the alveoli, prevents normal exchange of oxygen and carbon dioxide. As more alveoli collapse, less of the lung is ventilated, and hypoventilation occurs.

Signs and symptoms of hypoventilation include

- Mental status changes
- Dysrhythmias, and potential cardiac arrest. If untreated, the patient's status rapidly declines, leading to convulsions, unconsciousness, and death.

II- Hyperventilation:

- It is a state of ventilation in which the lungs remove carbon dioxide faster than it is produced by cellular metabolism. Severe anxiety, infection, drugs, or an acid-base imbalance induces hyperventilation.

Signs and symptoms of hyperventilation include

Rapid respirations,

Numbness and tingling of hands/feet,

Light-headedness,

Loss of consciousness.

III-Hypoxia:

- It is inadequate tissue oxygenation at the cellular level. It results from a deficiency in oxygen delivery or oxygen use at the cellular level. It is a life-threatening condition.
- **Hypoxemia**: refers to a decrease in the amount of arterial oxygen (o_2 in blood).

The clinical signs and symptoms of hypoxia include: -

- Apprehension, restlessness, inability to concentrate,
- Decreased level of consciousness, dizziness, and behavioral changes.
- The patient with hypoxia is unable to lie flat and appears both fatigued and agitated.
- Vital signs changes include an increased pulse rate and rate and depth of respiration.

IV- Cyanosis:

- blue discoloration of the skin and mucous membranes.
- Cyanosis ***caused by*** the presence of de-saturated hemoglobin in capillaries, is a late sign of hypoxia.
- **Types of cyanosis:**
- ***Central cyanosis***, observed in the tongue, soft palate, and conjunctiva of the eye where blood flow is high, indicates hypoxemia.
- ***Peripheral cyanosis***, seen in the extremities, nail beds, and earlobes, is often a result of vasoconstriction and stagnant blood flow.

Nursing management

- **a- Nursing History**
- A comprehensive nursing history relevant to oxygenation status should include data about current and past respiratory problems; lifestyle; presence of cough, sputum, or pain; medications for breathing; and presence of risk factors for impaired oxygenation status.
- **b- Physical Examination**
- In assessing a client's oxygenation status, the nurse uses all four physical examination techniques: inspection, palpation, percussion, and auscultation.
- **c- Diagnostic Studies**
- The primary care provider may order various diagnostic tests to assess respiratory status, function, and oxygenation. Included are sputum specimens, throat cultures, visualization procedures, venous and arterial blood specimens, and pulmonary function tests. Measurement of arterial blood gases is an important diagnostic procedure.

Diagnosing

- ***Ineffective Airway Clearance:*** inability to clear secretions or obstructions from the respiratory tract to maintain a clear airway.
- ***Ineffective Breathing Pattern***: inspiration and/or expiration that does not provide adequate ventilation.
- ***Impaired Gas Exchange***: excess or deficit in oxygenation and/or carbon dioxide elimination at the alveolar-capillary membrane.

Planning

- Maintain a patent airway.
- Improve comfort and ease of breathing.
- Maintain or improve pulmonary ventilation and oxygenation.
- Improve the ability to participate in physical activities.
- Prevent risks associated with oxygenation problems such as skin and tissue breakdown, syncope, acid–base imbalances, and feelings of hopelessness and social isolation.

Implementation

Nursing interventions to facilitate pulmonary ventilation may include:

- Ensuring a patent airway
- Positioning
- Encouraging deep breathing and coughing, Postural drainage, and percussion and vibration.
- Ensuring adequate hydration.
- Suctioning if need
- Administration of analgesics before deep breathing and coughing

General lines of management

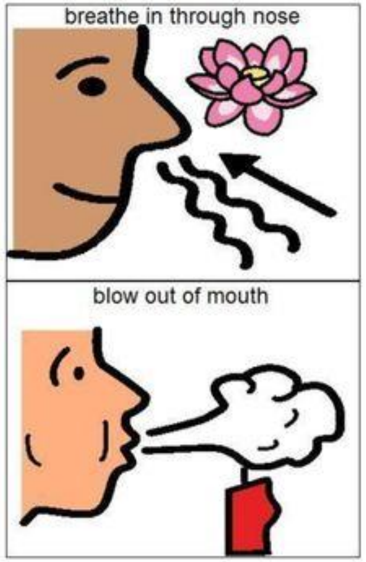
- Promoting Oxygenation
- Deep Breathing and Coughing
- Hydration
- Humidifiers
- Facilitating Respiratory Health
- Improving physical mobility
- Mobilizing secretions
- Medications
- Incentive Spirometer
- Chest Physiotherapy Therapy
- Oxygen Therapy

1. Promoting Oxygenation

- Changing position frequently, ambulating, and exercising usually maintain adequate ventilation and gas exchange.
- **Interventions by the nurse to maintain the normal respirations of clients include:**
  - Positioning the client to allow for maximum chest expansion
  - Encouraging or providing frequent changes in position
  - Encouraging deep breathing and coughing
  - Encouraging ambulation
  - Implementing measures that promote comfort, such as giving pain medications.

Promoting Oxygenation

- Positioning: Fowler's position
- Breathing techniques
  - Deep breathing
    - Incentive spirometry
  - Pursed-lip breathing
  - Diaphragmatic breathing
  - Nasal strips



- **2. Deep Breathing and Coughing:** The nurse can facilitate respiratory functioning by encouraging deep-breathing exercises and coughing to remove secretions from the airways. When coughing raises secretions high enough, the client expectorates (spit out) them.
- **3. Hydration:** Adequate hydration maintains the moisture of the respiratory mucous membranes.
- **4. Humidifiers:** are devices that add water vapor to inspired air. Room humidifiers provide cool mist to room air. Nebulizers are used to deliver humidity and medications.

5. Facilitating Respiratory Health

- 1. Exercise regularly—30 minutes, three to four times per week.
- 2. Do not smoke or use tobacco products.
- 3. Avoid secondhand smoke.
- 4. Reduce exposure to noxious fumes at home and at work.

6. Improving physical mobility: Maintaining physical and respiratory functioning becomes an important nursing intervention. The best position for maximum chest expansion is upright.

7. Medications: Several types of medications can be used for clients with oxygenation problems. *Bronchodilators*, anti-inflammatory drugs, expectorants, and cough suppressants are some medications that may be used to treat respiratory problems.

8. Mobilizing secretions

- Clearing respiratory secretions promotes patent airways and easier ventilation of the lungs, and prevents mucus stasis. Secretions are easier to cough up if they are thin, rather than thick and tenacious. Therefore, adequate fluid intake, humidification of inspired air, and possibly expectorant medication are important interventions in mobilizing secretions. Other medications that can be used before breathing exercises are inhaled bronchodilators or systemic bronchodilators . These medications allow the airways to relax and dilate so that the breathing and coughing exercises are more effective.

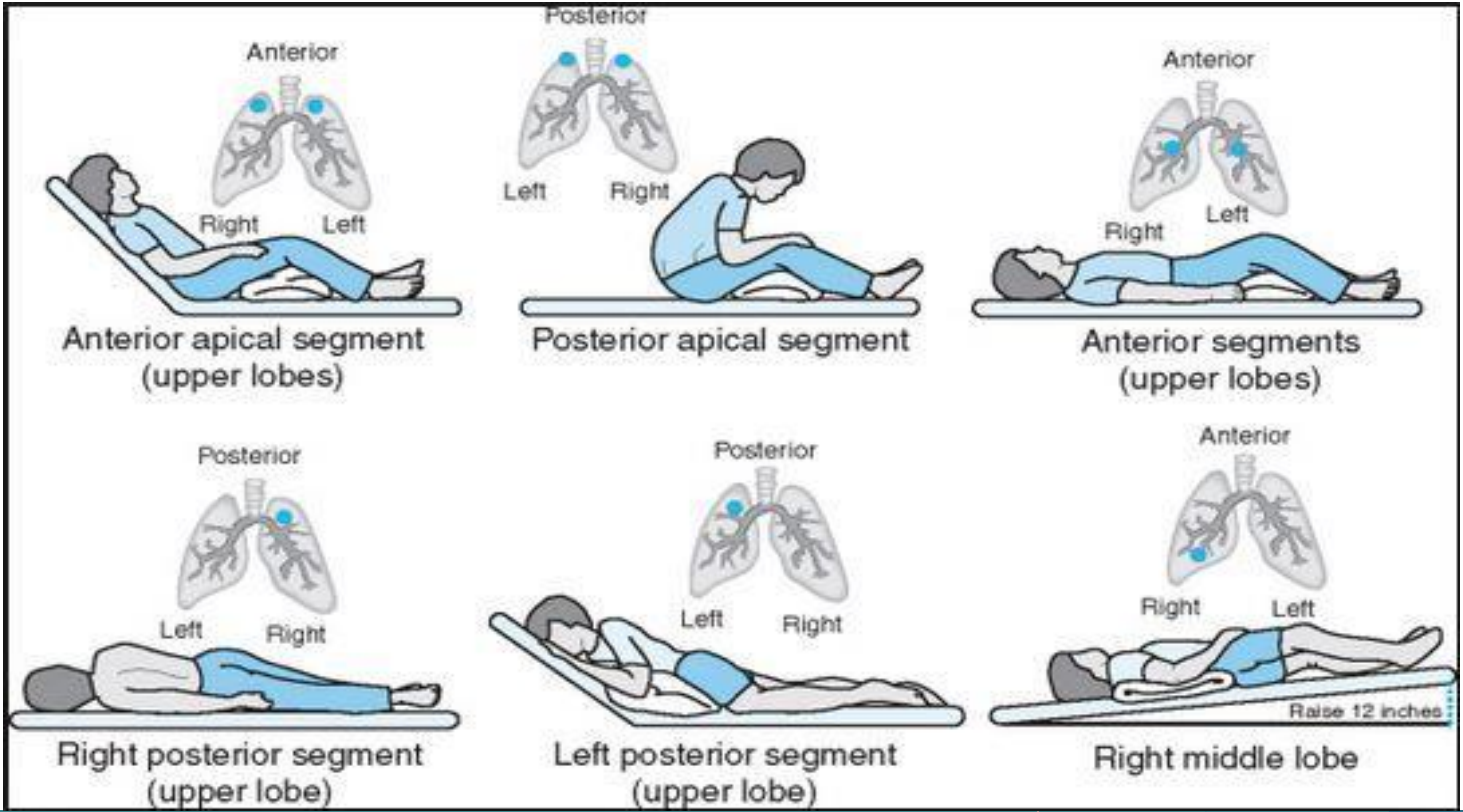
9. Incentive Spirometer:

- Incentive spirometers also referred to as *sustained maximal inspiration devices (SMIs)*, measure the flow of air inhaled through the mouthpiece and are used to:
- Improve pulmonary ventilation.
- Facilitate respiratory gaseous exchange.
- Loosen respiratory secretions.
- Expand collapsed alveoli.

10. Chest Physical Therapy

- Chest physical therapy (CPT) is another method to mobilize secretions and maintain airway patency and alveolar expansion. It may be combined with other interventions such as incentive spirometry, inhaled bronchodilators, and suctioning.

- **Chest physical therapy is composed of three techniques that may be used individually or in combination.** These techniques **percussion**, **vibration**, and **postural drainage** are performed by respiratory therapists or nurses.


Anterior
Right
Left
Anterior apical segment
(upper lobes)
Posterior
Left
Right
Posterior apical segment
Anterior
Right
Left
Anterior segments
(upper lobes)
Posterior
Left
Right
Right posterior segment
(upper lobe)
Posterior
Left
Right
Left posterior segment
(upper lobe)
Anterior
Right
Left
Raise 12 inches
Right middle lobe

11. Oxygen Therapy

- Some clients require oxygen therapy to keep a healthy level of tissue oxygenation. The goal of oxygen therapy is to prevent or relieve hypoxia.


Thank you